CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A data analytics project exploring customer purchasing patterns, product preferences, discount behavior, subscriptions and revenue contribution.

PROJECT REPORT

Tools and technologies: Python • Pandas • PostgreSQL • SQL • Power BI

3,900 purchase records	**18** source columns	**10** business questions

Project Overview

This project applies data analytics techniques to transform customer shopping records into meaningful business insights. The workflow covers data preparation, exploratory analysis, database integration, SQL-based business analysis and interactive dashboard development.

1. Executive Summary

The Customer Shopping Behavior Analysis project examines 3,900 purchase records across multiple product categories. The objective is to understand customer spending patterns, product preferences, discount usage, shipping choices and subscription behavior.

The analysis begins with Python and Pandas to inspect and prepare the dataset. Missing review ratings are handled using category-level medians, column names are standardized, and additional fields are created for age groups and purchase frequency. The cleaned data is then loaded into PostgreSQL, where SQL queries are used to answer ten business questions. Power BI brings the main metrics and comparisons together in an interactive dashboard.

Key Findings

- Male customers generated USD 157,890 in revenue, while female customers generated USD 75,191.
- Gloves had the highest average review rating among the five leading products, at 3.86.
- The average purchase amount was USD 60.48 for Express shipping and USD 58.46 for Standard shipping.
- Subscribers generated USD 62,645 in total revenue, compared with USD 170,436 from non-subscribers. Average spend was similar across the two groups.
- Customer segmentation identified 3,116 Loyal, 701 Returning and 83 New records.
- Young Adults contributed the highest revenue among the four age groups, at USD 62,143.

These findings provide a basis for examining customer engagement, promotional effectiveness, product positioning and retention opportunities.

2. Project Background and Objectives

Retail transaction data provides an opportunity to understand customer preferences and purchasing behavior. By organizing this information and applying analytical techniques, businesses can identify patterns that support decisions related to marketing, product strategy and customer experience.

2.1 Project Objectives

- Prepare the customer shopping dataset for reliable analysis.
- Identify and handle missing values in the Review Rating field.
- Create additional analytical fields for age groups and purchase frequency.
- Compare revenue and average purchase amounts across customer groups.
- Identify highly rated products and products with higher discount-applied rates.
- Analyze subscription status and previous-purchase behavior.
- Present key metrics through an interactive Power BI dashboard.
- Develop business recommendations based on the analytical findings.

2.2 Project Scope

The project covers data cleaning and feature engineering in Python, database integration with PostgreSQL, ten SQL-based business analyses and dashboard development in Power BI. The analysis focuses on descriptive insights from customer purchase records.

2.3 Business Value

The results can help identify customer groups contributing to revenue, products receiving stronger ratings, categories with higher sales activity and areas where discount or subscription strategies may need further evaluation.

3. Dataset Overview

The dataset contains 3,900 rows and 18 columns. It combines customer profile information with product, transaction and shopping-behavior attributes.

Data Area	Fields
Customer Profile	Age, Gender, Location and Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size and Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating and Shipping Type
Dataset Size	3,900 rows and 18 columns
Missing Values	37 missing values in Review Rating

3.1 Data Quality

The Review Rating column contained 37 missing values. These values were addressed during data preparation using the median rating within each product category. This approach retains the records while using a category-specific value for imputation.

3.2 Data Preparation Requirements

The dataset required consistent column naming, treatment of missing ratings, creation of additional analytical fields and a check for overlapping information between discount_applied and promo_code_used. These steps prepared the data for database storage and SQL analysis.

3.3 Data Considerations

The analysis uses the purchase amount field in USD. The dataset's observation period is not defined in the analysis. Customer-level measures should also distinguish between purchase records and unique customers, depending on the metric being calculated.

4. Data Preparation and Exploratory Data Analysis Using Python

Python was used to inspect, clean and transform the customer shopping dataset before it was loaded into PostgreSQL.

4.1 Data Loading

The dataset was imported into a Pandas DataFrame. Pandas provided the main environment for inspecting and transforming the records.

4.2 Initial Data Exploration

The first rows were reviewed using `head()`. The `info()` method was used to inspect column names, data types and non-null counts, while `describe(include='all')` generated summary statistics. Missing values were identified using `isnull().sum()`.

4.3 Missing-Value Treatment

The 37 missing values in Review Rating were filled using the median rating of the corresponding product category. A group-wise transformation was applied so that each missing rating was replaced with the median of its category. The missing-value count was checked again after imputation.

4.4 Column Standardization

Column names were converted to lowercase and spaces were replaced with underscores to create a consistent snake_case naming convention. The purchase amount column was renamed to `purchase_amount_usd` for easier reference in Python and SQL.

4.5 Age Group Creation

A new `age_group` field was created using `pd.qcut()` with four quantile-based groups. The labels used were Young Adult, Adult, Middle Aged and Senior. Since qcut divides observations according to the distribution of age values, the group boundaries are determined by the dataset rather than fixed age ranges.

4.6 Purchase Frequency Conversion

A new field named `purchase_frequency_days` was created by mapping the frequency categories to approximate day values.

Frequency Category	Mapped Days
Weekly	7
Fortnightly	14
Bi-Weekly	14
Monthly	30
Quarterly	90
Every 3 Months	90
Annually	365

These values represent the mapping used in the transformation. They are approximate day equivalents rather than durations calculated from transaction dates.

4.7 Redundant Column Check

The values in `discount_applied` and `promo_code_used` were compared to determine whether both columns contained identical information. After the equality check, `promo_code_used` was removed from the working DataFrame.

4.8 PostgreSQL Integration

The cleaned DataFrame was connected to PostgreSQL using SQLAlchemy and the PostgreSQL driver. The data was loaded into the `customer` table in the `customer_behavior` database. The table was replaced during the load operation to ensure that the database contained the latest prepared dataset.

5. End-to-End Analytical Workflow

The project follows a structured process from raw data preparation to database analysis and dashboard reporting.

Stage	Activity	Output
1. Data Loading	Import the CSV file into a Pandas DataFrame.	Working DataFrame
2. Data Inspection	Review rows, structure, data types, summary statistics and missing values.	Understanding of dataset quality
3. Data Cleaning	Impute missing Review Rating values using category medians.	Prepared rating field
4. Standardization	Convert column names to snake_case and rename the purchase amount field.	Consistent field names
5. Feature Engineering	Create quartile-based age_group and purchase_frequency_days fields.	Additional analytical attributes
6. Validation	Compare discount_applied and promo_code_used, then remove the redundant field.	Simplified dataset
7. Database Integration	Load the cleaned DataFrame into the PostgreSQL customer table.	SQL-ready dataset
8. Business Analysis	Execute SQL queries and present key findings in Power BI.	Business insights and dashboard

6. Business Analysis Using PostgreSQL

PostgreSQL was used to analyze customer purchases and answer ten business questions. The queries use filtering, aggregation, conditional logic and ranking to summarize the dataset.

6.1 Revenue by Gender

Business Question: What is the total revenue generated by male and female customers?

Method: Records were grouped by gender and the purchase amount was summed for each group using GROUP BY and SUM().

Gender	Revenue (USD)
Male	157,890
Female	75,191

Business Insight: Male customers contributed a higher share of total revenue in the dataset. This comparison can be used to explore differences in customer count, average spend and product preferences across gender groups.

6.2 Discount Users with Above-Average Purchases

Business Question: Which customers used a discount but still spent at least the average purchase amount?

Method: The query filters records where discount_applied is Yes and compares each purchase amount with the overall average calculated through a scalar subquery. The >= condition includes purchases equal to the average as well as purchases above it.

Result: The query returned 839 qualifying records.

Business Insight: Discounted purchases are not necessarily low-value transactions. Some discounted records meet or exceed the overall average purchase amount. This provides an opportunity to investigate whether discounts are being used by customers who may already have a higher spending level.

6.3 Top Five Products by Average Review Rating

Business Question: Which five products have the highest average review ratings?

Method: Records were grouped by item_purchased, the average review rating was calculated, and the results were sorted in descending order.

Product	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Business Insight: Gloves, Sandals and Boots are among the highest-rated products. These products can be considered for merchandising and promotional visibility, while also reviewing the number of ratings behind each average.

6.4 Average Purchase Amount by Shipping Type

Business Question: How does the average purchase amount compare between Standard and Express shipping?

Method: The analysis filters Standard and Express shipping records, groups them by shipping type and calculates the average purchase amount.

6. Business Analysis Using PostgreSQL (Continued)

Shipping Type	Average Purchase Amount (USD)
Standard	58.46
Express	60.48

Business Insight: Express-shipping purchases have a higher average purchase amount by USD 2.02. Further analysis of basket size and customer preferences can help explain this difference.

6.5 Subscribers versus Non-Subscribers

Business Question: Do subscribed customers spend more? The analysis compares record count, average spend and total revenue.

Method: The data is grouped by subscription_status and analyzed using COUNT(), AVG() and SUM().

Status	Record Count	Average Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Business Insight: Non-subscribers contribute higher total revenue because they represent a larger share of the records. Average spend is similar across the two groups, with non-subscribers slightly higher in this analysis. Subscription engagement and conversion can be examined further.

6.6 Products with the Highest Discount-Applied Rate

Business Question: Which five products have the highest percentage of purchases with discounts applied?

Method: A CASE expression assigns 1 to discounted purchases and 0 to other purchases. The indicators are summed, divided by the total number of records for each product and converted into a percentage.

Product	Discount-Applied Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Business Insight: Hats have the highest discount-applied rate among the listed products. These products can be reviewed alongside gross margin, discount depth and full-price sales to understand promotional effectiveness.

6.7 Customer Segmentation by Previous Purchases

Business Question: How many customers fall into New, Returning and Loyal segments?

Method: A Common Table Expression (CTE) and CASE statement classify records: previous_purchases = 1 as New, values from 2 to 10 as Returning, and all other values as Loyal.

Customer Segment	Record Count
Loyal	3,116
Returning	701
New	83

Business Insight: Loyal is the largest segment under the defined purchase-history rules. The segment thresholds provide a starting point for customer lifecycle analysis and can be refined for operational use.

6. Business Analysis Using PostgreSQL (Continued)

6.8 Top Three Products within Each Category

Business Question: What are the three most purchased products within each category?

Method: Purchase counts are aggregated by category and item. ROW_NUMBER() with PARTITION BY category ranks products separately within each category.

Category	Ranked Products
Accessories	Jewelry (171), Sunglasses (161), Belt (161)
Clothing	Blouse (171), Pants (171), Shirt (169)
Footwear	Sandals (160), Shoes (150), Sneakers (145)
Outerwear	Jacket (163), Coat (161)

Business Insight: The results identify leading products within each category. The Outerwear output contains two displayed products. Ranking ties can be resolved consistently by adding a secondary sort condition to the window function.

6.9 Repeat Buyers and Subscription Status

Business Question: Among customers with more than five previous purchases, how many are subscribers and non-subscribers?

Method: Records are filtered using previous_purchases > 5, grouped by subscription_status and counted.

Subscription Status	Repeat-Buyer Records
No	2,518
Yes	958

Business Insight: More repeat-buyer records are marked as non-subscribers. To compare subscription likelihood accurately, subscription rates should be calculated using the total number of repeat buyers and the relevant comparison groups as denominators.

6. Business Analysis Using PostgreSQL (Continued)

6.10 Revenue Contribution by Age Group

Business Question: What is the revenue contribution of each age group?

Method: Records are grouped by age_group, purchase amounts are summed and the groups are sorted by total revenue.

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Business Insight: Young Adults contribute the highest revenue, followed by Middle-aged customers. These groups can be studied further through average purchase amount, product preferences and purchase frequency.

6.11 SQL Concepts Applied

SQL Concept	Application
GROUP BY	Groups records by gender, shipping type, subscription status, product or category.
SUM()	Calculates total revenue and conditional totals.
AVG()	Calculates average purchase amounts and product ratings.
COUNT()	Counts records within each group.
WHERE	Filters discounted purchases and repeat buyers.
Scalar Subquery	Calculates the overall average purchase amount for comparison.
CASE	Creates customer segments and conditional indicators.
CTE	Organizes intermediate query results.
ROW_NUMBER() / PARTITION BY	Ranks products within each category.
ORDER BY / LIMIT	Sorts results and returns the required top records.

7. Power BI Dashboard

An interactive Power BI dashboard was developed using a direct connection to the customer data stored in PostgreSQL. The dashboard combines key performance indicators, customer distribution, category-level comparisons and age-group analysis in a single report.

7.1 KPI Measures

Three measures were created in Power BI to summarize the main customer and purchase metrics.

Measure	Dashboard Value	Purpose
Number of Customer	3,900	Displays the customer count represented by the measure.
Average Purchase Amount	$59.76	Shows the average purchase amount in the current filter context.
Average Review Rating	3.75	Displays the average review rating in the current filter context.

7.2 Dashboard Visuals

Visual	Description
Subscription Status	A doughnut chart showing the distribution of subscribers and non-subscribers. The displayed split is approximately 27% Yes and 73% No.
Revenue by Category	A column chart comparing revenue across Clothing, Accessories, Footwear and Outerwear.
Sales by Category	A column chart comparing sales volume across the four product categories.
Revenue by Age Group	A horizontal bar chart comparing revenue across Young Adult, Middle Aged, Adult and Senior groups.
Sales by Age Group	A horizontal bar chart comparing sales volume across the age groups.

7.3 Interactive Slicers

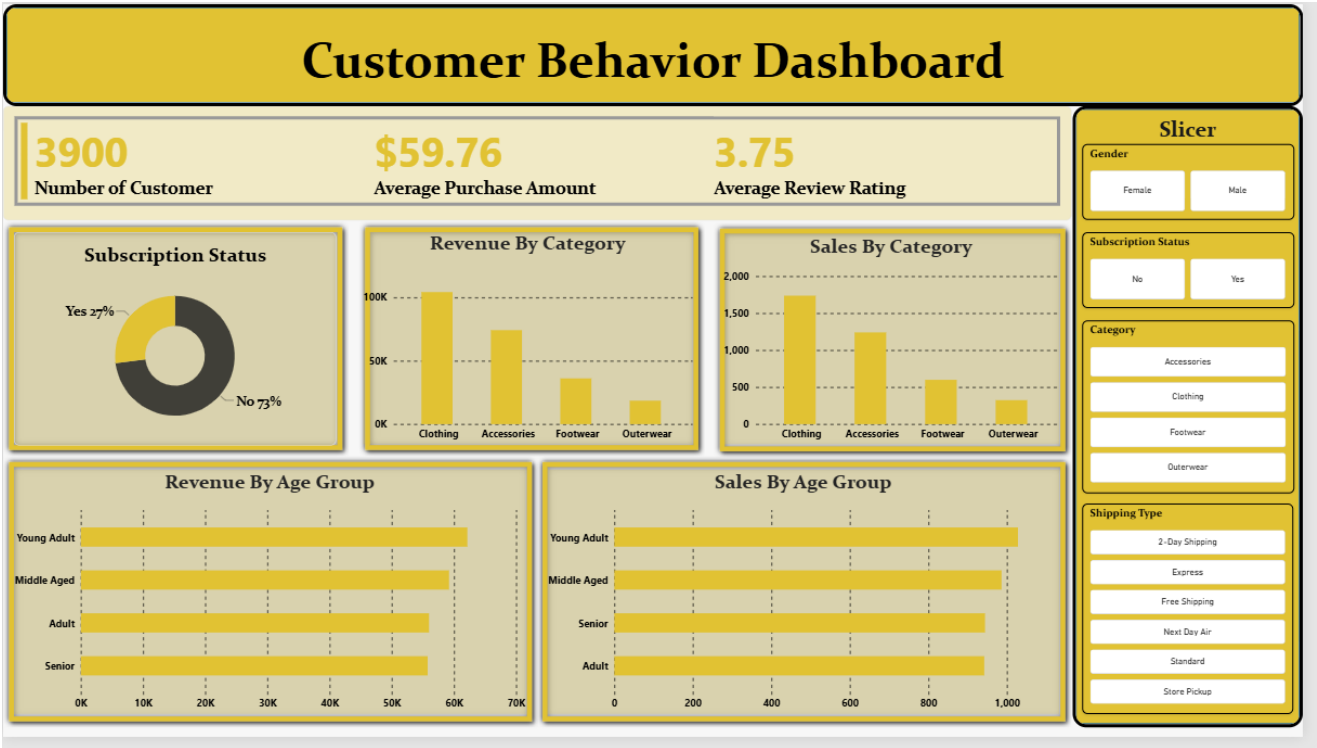
The dashboard includes slicers for Gender (Female and Male), Subscription Status (No and Yes), Category (Accessories, Clothing, Footwear and Outerwear), and Shipping Type (2-Day Shipping, Express, Free Shipping, Next Day Air, Standard and Store Pickup). These filters allow users to explore the metrics for selected customer groups, product categories and shipping preferences.

7.4 Dashboard Design and Business Use

Revenue and sales-volume visuals are presented separately so that performance can be evaluated from more than one perspective. The subscription chart highlights the customer mix, while the slicers make it possible to compare patterns across demographic, subscription, product and shipping dimensions.

Power BI Dashboard

Customer Behavior Dashboard — PostgreSQL-connected interactive report.



The dashboard provides a consolidated view of customer metrics, category performance, age-group contribution and subscription distribution. Interactive slicers support focused comparisons across the available dimensions.

8. Business Recommendations

The analysis highlights several opportunities for improving customer engagement, evaluating promotions and strengthening product and category strategies.

1. Improve Subscription Engagement

Average spend is similar for subscribers and non-subscribers, while non-subscribers contribute higher total revenue. Review the subscription benefits and customer journey, and test clearer value communication to improve sign-up and renewal.

2. Evaluate Product-Level Discount Strategy

Products with higher discount-applied rates should be reviewed alongside gross margin, discount depth and full-price sales. This will help determine whether promotions generate incremental demand while protecting profitability.

3. Promote Highly Rated Products

Products such as Gloves, Sandals and Boots have the highest average ratings. Consider featuring these products in relevant merchandising placements and campaigns, while monitoring review volume and stock availability.

4. Review Shipping Preferences

Express-shipping purchases have a higher average purchase amount than Standard-shipping purchases. Analyze basket size, customer mix and delivery preferences to understand this difference.

5. Strengthen Customer Retention

Use the New, Returning and Loyal segments to guide customer lifecycle communication. Encourage repeat purchases among returning customers and develop relevant engagement strategies for loyal customers.

6. Use Age-Group Insights in Marketing

Young Adults contribute the highest revenue among the age groups. Explore their category preferences, average spend and purchase frequency before allocating marketing resources.

9. Limitations and Further Analysis

- The dataset's observation period is not defined, so time-based trends cannot be evaluated.
- The 37 missing Review Rating values were imputed using category medians. Imputed values may influence product-level average ratings.
- Age groups are created using quartiles, so their boundaries depend on the distribution of ages in the dataset.
- Purchase-frequency day values are approximate mappings from category labels rather than elapsed days calculated from transaction dates.
- The promo_code_used field was removed after checking its equality with discount_applied. Separate promotion-code analysis would require distinct information in that field.
- COUNT(customer_id) counts records. If a customer can appear in multiple purchase records, distinct customer counts should be calculated separately.
- The repeat-buyer analysis compares counts by subscription status but does not calculate subscription rates or likelihood.
- Discount-applied rates do not include gross margin or discount depth, so promotional profitability cannot be determined from these results alone.
- The Outerwear category output displays two products rather than three.
- The customer segmentation rules classify previous_purchases values outside 1–10 as Loyal; the thresholds should be reviewed before operational use.

Recommended Further Analysis

- Calculate distinct customer counts and customer-level revenue measures.
- Compare subscription rates across customer segments using clear denominators.
- Measure customer retention and repeat-purchase rates if transaction dates are available.
- Include gross margin and discount depth in promotional analysis.
- Review rating distributions and review counts alongside average ratings.
- Validate Power BI measures against SQL calculations and confirm filter behavior.

10. Conclusion

This project demonstrates an end-to-end approach to customer shopping behavior analysis. Python and Pandas were used to inspect, clean and transform the dataset. PostgreSQL provided the environment for structured business analysis, while SQL queries were used to investigate revenue, product ratings, discounts, shipping, subscriptions, customer segments and age-group contribution.

The Power BI dashboard brings the main metrics and comparisons together in an interactive format. The combined analysis provides a foundation for business decisions related to customer engagement, promotional strategy, product positioning and retention. Further analysis using customer-level measures, margin data and transaction dates can strengthen these insights.

Appendix A. SQL Business Questions

No.	Business Question	SQL Techniques
1	Revenue by gender	GROUP BY, SUM
2	Discount users with purchases at or above average	WHERE, subquery, AVG
3	Top five products by average rating	GROUP BY, AVG, ORDER BY, LIMIT
4	Average purchase amount by shipping type	WHERE, GROUP BY, AVG
5	Subscriber versus non-subscriber spend and revenue	COUNT, AVG, SUM
6	Products with highest discount-applied rate	CASE, SUM, COUNT
7	Customer segmentation by previous purchases	CTE, CASE, COUNT
8	Top three products within each category	CTE, ROW_NUMBER, PARTITION BY
9	Repeat buyers by subscription status	WHERE, GROUP BY, COUNT
10	Revenue by age group	GROUP BY, SUM, ORDER BY

Appendix B. Key Terms

Term	Definition
Category-Level Median Imputation	Replacing missing review ratings with the median rating from the same product category.
Quartile Binning	Dividing age observations into four groups based on their distribution.
Feature Engineering	Creating additional analytical fields such as age_group and purchase_frequency_days.
Discount-Applied Rate	The percentage of purchase records for an item where a discount was applied.
CTE	A named temporary result created using WITH and referenced by a subsequent query.
Window Function	A function that calculates values across related rows, such as ranking products within each category.
Descriptive Analysis	Analysis that summarizes observed data without establishing cause and effect.